EXPERIMENT-7

```
clc;
clear;
close all;
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal
%% Message Signal Parameters
Am = 2;              % Message amplitude (V)
fm = 500;             % Message frequency (Hz)
%% Carrier Signal Parameters
Ac = 1;              % Carrier amplitude (V)
fc = 10000;           % Carrier frequency (Hz)
%% FM Parameters
kf = 250;             % Frequency sensitivity (Hz/V)
fs = 100000;          % Sampling frequency (Hz)
T = 0.01;             % Simulation duration (s)
%% Time Vector
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation
delta_f = kf*Am;
%% Modulation Index
beta = delta_f/fm;
%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
%% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n', fc);
fprintf('Message Frequency (fm) = %.0f Hz\n', fm);
```

```
fprintf('Message Amplitude (Am) = %.0f V\n', Am);
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n', kf);
fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n', delta_f);
fprintf('Modulation Index (Beta) = %.2f\n', beta);
%% Verification of Modulation Index
beta_verify = delta_f/fm;
fprintf('Verified Modulation Index = %.2f\n', beta_verify);
%% Verification Check
tolerance = 1e-6;
if abs(beta - beta_verify) < tolerance
    fprintf('\nVerification Successful\n');
else
    fprintf('\nVerification Failed\n');
end
%% Plotting Signals
figure;
subplot(2,1,1);
plot(t, m, 'LineWidth', 1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(2,1,2);
plot(t, fm_signal, 'LineWidth', 1);
grid on;
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
```

Editor

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```
clc;
clear;
close all;

% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal

%% Message Signal Parameters
Am = 2;                 % Message amplitude (V)
fm = 500;               % Message frequency (Hz)

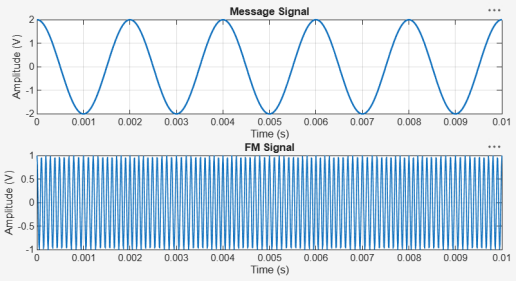
%% Carrier Signal Parameters
Ac = 1;                 % Carrier amplitude (V)
fc = 10000;             % Carrier frequency (Hz)

%% FM Parameters
kf = 250;               % Frequency sensitivity (Hz/V)
fs = 100000;            % Sampling frequency (Hz)
T = 0.01;               % Simulation duration (s)

```

Figures

Figure 1



Command Window

```
Frequency Deviation (Delta f) = 500.00 Hz
Modulation Index (Beta) = 1.00
Verified Modulation Index = 1.00

Verification Successful
>>
```